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Review Article

Efficacy of Laser Acupuncture for Treatment of Chronic Low Back Pain: A Systematic Review and Meta-Analysis

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ABSTRACT

Objective: To summarize and systematically analyze the efficacy of laser acupuncture (LA) interventions in reducing pain scores in patients suffering from chronic low back pain (LBP).

Methods: PubMed, EMBASE, and Scopus databases were searched for randomized controlled trials, published in peer-reviewed journals, and reporting LA interventions in patients with chronic LBP. All included studies had a comparison group of patients, receiving placebo treatment, sham intervention, conventional therapy, or no treatment. The outcome of interest was the pain intensity score. Pooled effect estimates were calculated using random-effects models and reported as weighted mean difference (WMD) with 95% confidence intervals (CI).

Results: A total of 20 studies were included. Compared to the control group, patients who underwent LA experienced a significant reduction in reported pain scores immediately after completing the treatment (WMD −1.14, 95% CI: −1.68 to −0.61). High dose of LA was associated with a more significant decrease in the pain scores (WMD −1.40, 95% CI: −1.94 to −0.85; $N = 15$, $I^2 = 81.0\%$). However, reported pain scores of patients who received LA were statistically similar to those of the control group at short-term (4-8 weeks after the treatment) and long-term (12 months) follow-ups.

Conclusions: In patients with chronic LBP, LA may help in alleviating pain immediately after the treatment. However, this effect does not appear to be sustained on later follow-up assessments. Consequently, patients should be informed about the potential limitations of the treatment in providing lasting pain relief.

Low back pain (LBP) is a prevalent and debilitating condition, imposing a significant burden on both patients and the healthcare system. In 2019, an estimated 224 million cases of LBP were reported worldwide (Wang et al., 2022). In the same year, LBP-related disability-adjusted life years amounted to a substantial 64 million, indicating the profound toll it takes on the quality of life and functional capacity of those affected (Wang et al., 2022). LBP is a multifaceted health issue, involving a complex interplay of musculoskeletal and psychosocial factors (Corrêa et al., 2022; Stevans et al., 2021). When no specific underlying pathological cause can be identified, the condition is categorized as nonspecific LBP (NSLBP) and is characterized by pain and discomfort localized in the lumbosacral region for a duration exceeding 3 months (Chou et al., 2007; Maher et al., 2017). This chronic nature leads to functional limitations, disability, and a notable impact on both daily activities and the overall quality of life (Maher et al., 2017).

Traditional treatment approaches to LBP include pharmacological agents, such as nonsteroidal anti-inflammatory drugs, antidepressants, and physical and psychosocial interventions (George et al., 2021; Mauck et al., 2022). However, the effectiveness of these treatment approaches is limited, and there is a need for developing alternative therapies. Recently, laser acupuncture (LA) has gained increasing attention as a promising and noninvasive method for managing LBP. While acupuncture has been used as a complementary and alternative medicine approach for NSLBP, the recommendations for its use in treatment guidelines remain inconsistent.

Acupuncture is a therapeutic technique, rooted in traditional Chinese medicine, that is based on stimulation of specific points on the body using fine needles to rebalance the body's energy flow (Zhu, 2014). LA is a modern adaptation of this ancient practice, that uses low-level laser devices to stimulate the acupoints, eliminating the need for needle insertion. The use of lasers in acupuncture

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presents potential advantages, such as reducing patient discomfort and eliminating the risk of infection, making it an attractive option for those seeking alternative treatments. LA has been predominantly employed in the management of arthralgia and musculoskeletal pain.

While there are some studies exploring its potential efficacy for chronic LBP (Hung et al., 2021), the effectiveness of LA for chronic LBP remains unclear. A Cochrane report published in year 2008 concluded that there is still not enough data to draw definitive conclusions regarding the clinical impact of laser therapy for LBP (Yousefi-Nooraie et al., 2008). The two existing meta-analyses of the studies, assessing the efficiency of LA in treating LBP suggested that LA was associated with short-term improvement in pain in LBP patients (Chen et al., 2022; Glazov et al., 2016).

The current meta-analysis aims to provide an update of existing evidence from randomized controlled trials (RCTs) and systematically analyze the efficacy of LA interventions in reducing pain scores in LBP patients. Our results will not only provide clinicians and patients with valuable insights into the benefits and limitations of LA but also inform future research directions in this field. The findings will be crucial for informing the practice of the nursing personnel as well, as they are the frontline healthcare professionals actively involved in pain management. It will empower nurses to offer informed guidance to patients, collaborate effectively with other healthcare professionals, and contribute to a holistic approach to pain management.

Methodology

Identification of Relevant Studies

A comprehensive and systematic search of PubMed, Embase, and Scopus was done with the primary aim to identify studies, published up until June 30, 2023, and related to the treatment of lower back pain using LA. A combination of keywords included (lower back pain OR low backache OR chronic backache OR nonspecific back pain) AND (LA OR low-level laser OR laser therapy) AND (pain score OR pain relief OR clinical outcomes OR pain severity). Manual searches of reference lists and pertinent review articles were also done to identify any additional relevant studies.

The two study authors, along with the lead author, were involved in the design of inclusion and exclusion criteria.

Inclusion criteria were as follows: (1) Only RCTs were included to ensure the inclusion of high-quality evidence; (2) adult (18 years and older) patients clinically diagnosed with chronic LBP; (3) the described intervention focused on LA, low-level laser therapy, or similar laser-based treatments; (4) studies having a comparison group: patients receiving placebo treatment, sham intervention, conventional therapy, or no treatment; (5) relevant outcome measures reported include pain intensity scores and pain relief.

Exclusion criteria were as follows: (1) Studies not in English; (2) nonrandomized trials, observational studies, case reports, editorials, conference abstracts, and letters; (3) studies involving pediatric populations or patients with acute LBP; (4) studies that were published prior to the year 2000; (5) studies that examined the effect of non-LA techniques, other unspecified forms of laser therapy, or combined therapies where the effect of LA could not be isolated; (6) studies with insufficient data or incomplete outcome reporting that hindered data extraction. In cases of the same study published in multiple articles, the most comprehensive or recent version was considered for inclusion. Duplicates were removed from the initial list of studies that were obtained from the implemented search strategy. Two authors independently conducted a review of the remaining studies. The titles and abstracts of each study were screened for their potential relevance to the research question. Studies that met the predefined criteria were shortlisted for further evaluation. In the subsequent stage, full texts of the selected studies were assessed to determine their eligibility for inclusion in the meta-analysis. Any discrepancies or disagreements regarding the inclusion of a study were addressed through extensive discussions among the study authors, ensuring a consensus-based decision-making process. Study selection was conducted in adherence to the PRISMA guidelines, which are designed to enhance the transparency and quality of systematic reviews and meta-analyses (PRISMA, n.d.). The study protocol was registered in PROSPERO (CRD42023447897), a prospective register for systematic reviews.

Data extraction, Quality Assessment, and Statistical Analysis

The extraction of data from the final set of studies was carried out independently by two authors using a standardized data extraction form. Any cases of disparities were resolved by comprehensive discussions. The quality of the included studies was assessed by the Cochrane risk of bias assessment tool for RCTs (RoB 2.0) (Sterne et al., 2019). The primary outcome for the review was the pain score. All the authors collectively determined the most appropriate statistical methods. For our analysis, the pooled effect sizes were reported as weighted mean difference (WMD) with 95% confidence intervals (CI). Since there may be variations in the baseline characteristics of the included studies, we made a predetermined decision to utilize a random-effects model for all analyses. Heterogeneity was calculated by I^2 statistics. High heterogeneity was considered when I^2 value exceed 40% (Higgins et al., 2023). We did a subgroup analysis based on laser dose, that is, low dose ($<$3 Joule/point) and high dose ($\geq$3 Joule/point). To understand whether differences in the number of sessions of laser therapy contributed to the high heterogeneity observed, for both high and low-dose laser categories, we further conducted a subgroup analysis based on the number of sessions of therapy received ($\leq$10 and $>$10). This was done only among studies reporting the outcome immediately after treatment completion as the number of studies for conducting subgroup analysis was more. Assessment of publication bias was done using Egger's test (Egger et al., 1997). $p < .05$ indicated statistical significance.

Results

The initial search identified 312 studies (Fig. 1). After the removal of 69 duplicates, and a thorough screening of the titles and abstracts of the remaining 243 studies, an additional 188 studies were removed as not meeting the predefined criteria. After the detailed full paper review, 35 studies that did not align with our specific eligibility criteria were excluded. Finally, 20 studies were selected for the analysis (Fig. 1) (Abdelbasset, Nambi, Alsubaie, et al., 2020; Abdelbasset, Nambi, Elsayed, et al., 2020; Alayat et al., 2014; Ammar, 2015; Ay et al., 2010; Cheng et al., 2023; Choi et al., 2017; de Carvalho et al., 2016; Djavid et al., 2007; Glazov et al., 2009, 2014; Gur et al., 2003; Hsieh & Lee, 2014; Kim et al., 2022; Lin et al., 2012, 2017; Shin et al., 2015; Tantawy et al., 2019; Vallone et al., 2014; Yang et al., 2023). The details of the included studies are presented in Table 1.

The risk of bias in the studies is summarized in Figure 2. Most studies (16/20, 80%) had a low risk of bias for the overall randomization process; 75% of studies (15/20), had a low risk of bias for deviation from intended intervention; 70% of studies (14/20) reported a low risk of bias for missing outcome data. A total of 75% (15/20) of studies had a low risk of bias for the measurement of outcome; and 60% (12/20) of studies had a low risk of bias for the selection of the reported results. We found four trials with a pos-

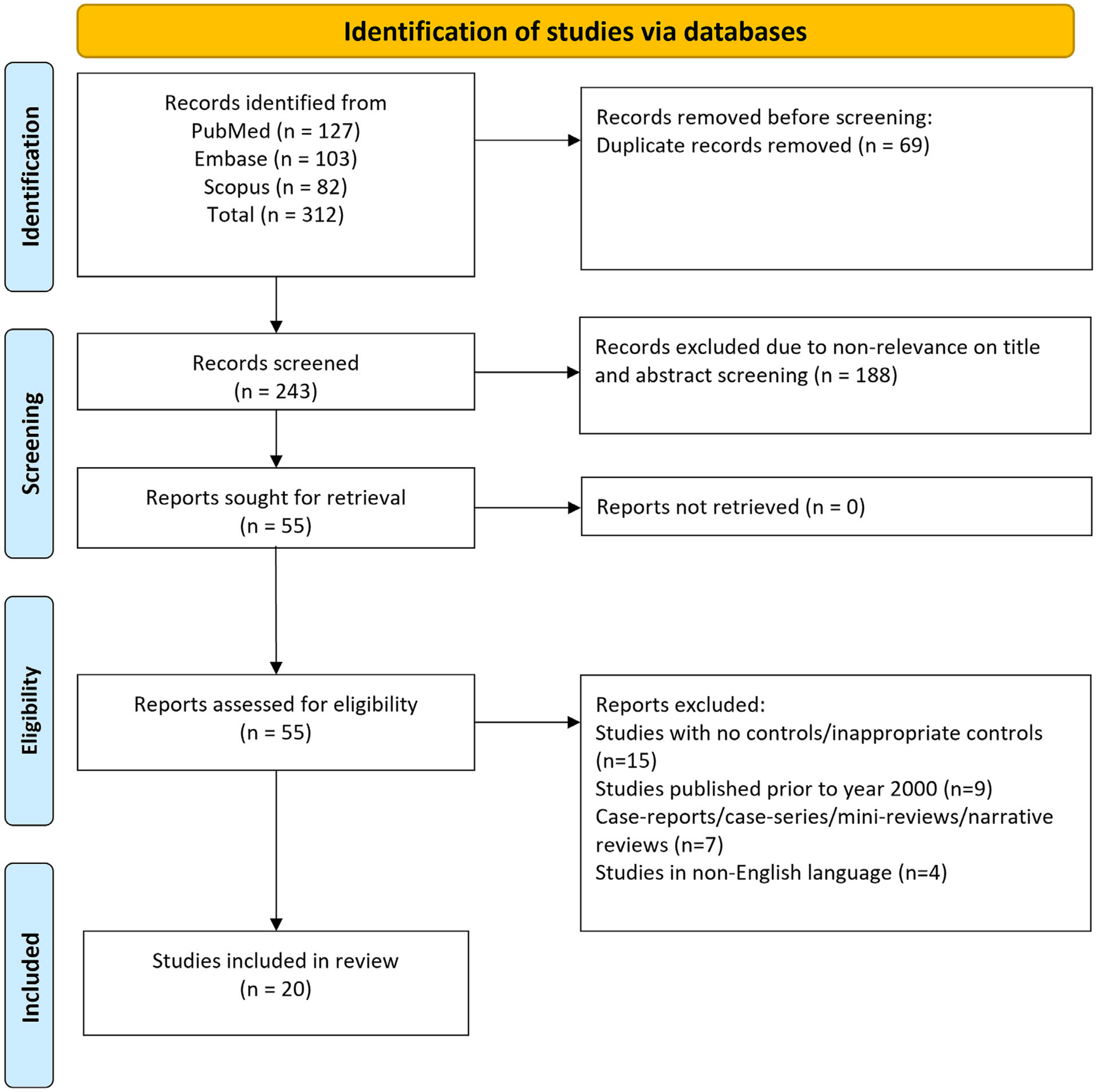

Figure 1. Selection process of studies included in the review.

sible higher risk of bias (Ammar, 2015; Choi et al., 2017; Lin et al., 2017; Shin et al., 2015).

All the included RCTs used the visual analog scale (VAS) for reporting of pain scores. Five studies were done in Taiwan, four in Saudi Arabia, and three in the Republic of Korea. Two studies each were done in Australia and Turkey. One study each was done in Egypt, Iran, Italy, and Brazil. In 13 studies, the control group received sham/placebo laser treatment. Exercise therapy was reported as a co-intervention (i.e., offered to both experimental and control groups) in 12 studies, soft cupping was a co-intervention in three studies, and hot packs- in two studies (Table 1). There were 13 studies that used a high laser dose (≥3 Joule/point), five studies employed a low dose (<3 Joule/point) and two studies tested both high and low doses. The duration of laser therapy in the included studies ranged from 1 to 12 weeks (mean of 4.55 weeks) and the number of sessions ranged from 3 to 24 (mean of 11.4 sessions).

Effect of LA on Pain Scores

Assessment immediately after treatment completion

Compared to the control group, patients who received LA had a significant decrease in the reported pain score when assessed immediately after the completion of the treatment (WMD −1.14, 95% CI: −1.68 to −0.61) (Fig. 3). The effect was significant only when studies using high dose were pooled (WMD −1.40, 95% CI: −1.94 to −0.85; $N = 15$, $I^2 = 81.0\%$). No significant effect was detected by pooling the studies that used the low-dose LA (WMD −0.71, 95% CI: −1.52 to 0.11; $N = 7$, $I^2 = 90.1\%$). There was no evidence of publication bias according to the egger's test ($p = .48$ for combined analysis; $p = .39$ for high dose and $p = .22$ for low dose). Further subgroup analysis based on the number of sessions of laser therapy confirmed high heterogeneity (Supplementary Figures 1 and 2). Also, within both the "high dose" and "low dose" groups, pa-

Figure 2. Quality assessment of the included RCTs using the Cochrane risk of bias tool (RoB 2).

tients receiving a higher number of sessions (>10) reported a substantial decrease in pain score compared to patients receiving ≤10 sessions (Supplementary Figures 1 and 2).

Pain scores at short-term and long-term follow-up

Patients receiving LA and the control group had comparable reported pain scores when assessed 4-8 weeks after completing the treatment (i.e., short-term follow-up) (WMD −1.00, 95% CI: −2.07 to 0.08) and at 12 months following the procedure (long-term follow-up) (WMD 0.44, 95% CI: −0.22 to 1.10) (Figs. 4 and 5). For the short-term follow-up, the effect was significant only when studies using high-dose LA were pooled (WMD −1.45, 95% CI: −2.30 to −0.60; $N = 3$, $I^2 = 35.3\%$) (Fig. 4). There was no evidence of publication bias according to the egger's test ($p > .05$) for the assessment of pain scores at both time points.

Discussion

This study aimed to evaluate the efficacy of LA as a treatment for LBP based on available evidence from various RCTs. Our results suggest that LA is associated with an immediate small decrease in pain scores after the treatment but does not show sustained effectiveness on later follow-ups. This finding indicates that LA may offer some short-term pain relief for patients suffering from LBP. However, it is important to interpret our results with caution, as the effect size was relatively small.

Our findings are similar to the previous reviews on this issue. Glazov et al. (2016) in their review of 15 studies found that high dose (≥3 Joule/point) LA was associated with a significant pain reduction of up to WMD −1.40 (95% CI −1.91 to −0.88) at immediate and short-term follow-up (Glazov et al., 2016).

Table 1
Characteristics of the Included Studies

Author (Publication Year) and Place of Study	Subject Characteristics	Sample Size	Details of Laser Acupuncture	Co-Intervention/ Control Group
Alayat et al. (2014) Saudi Arabia	Mean age of around 33 years and mean duration of back pain of 14 months	I (intervention): 28 C (control): 24	Laser diode: Nd:YAG laser Pulse mode: pulsed wavelength (nm): 1,064 spot size (cm^2): 0.2 dose/point (J): 25 energy density (J/cm^2): 0.61 power density (W/cm^2): 8.9 treatment time: 4 weeks (12 sessions)	Co-intervention: exercise control group also received placebo (sham) laser
Ay et al. (2010) Turkey	Mean age of around 54 years and mean duration of back pain of 50 months	I: 20 C: 20	Laser diode: GaAlA Pulse mode: pulsed wavelength (nm): 805 spot size (cm^2): 0.07 dose/point (J): 2.8 energy density (J/cm^2): 40 power density (W/cm^2): 1.4 treatment time: 3 weeks (15 sessions)	Co-intervention: hot packs control group also received placebo (sham) laser
Djavid et al. (2007) Iran	Mean age of around 37 years and mean duration of back pain of 25 months	I: 19 C: 18	Laser diode: GaAlA Pulse mode: continuous wavelength (nm): 810 spot size (cm^2): 0.22 dose/point (J): ~7.5 energy density (J/cm^2): 27 power density (W/cm^2): 8.2 treatment time: 6 weeks (12 sessions)	Co-intervention: exercise control group also received placebo (sham) laser
Glazov et al. (2009) Australia	Median age of around 51 years and mean duration of back pain of 11 years	I:45 C:45	Laser diode: GaAlA Pulse mode: continuous wavelength (nm): 830 spot size (cm^2): 0.2 dose/point (J): 0.2 energy density (J/cm^2): 1 power density (W/cm^2): 0.05 treatment time: 10 weeks (10 sessions)	Co-intervention: exercise control group also received placebo (sham) laser
Glazov et al. (2014) Australia	Mean age of around 54 years and mean duration of back pain of 13 years	I:48 (each in high and low-dose group C:48	Laser diode: GaAlA Pulse mode: continuous wavelength (nm): 830 spot size (cm^2): 0.2 dose/point (J): 0.2 energy density (J/cm^2): 1 (in low dose) and 4 (in high dose) power density (W/cm^2): 0.1 treatment time: 8 weeks (eight sessions)	Co-intervention: exercise control group also received placebo (sham) laser
Lin et al. (2012)Taiwan	Mean age of around 63 years and mean BMI of around 26 kg/m^2	I:21 C:21	Laser diode: LA-400 Pulse mode: pulsed wavelength (nm): 808 spot size (cm^2): 0.8 dose/point (J): 12 energy density (J/cm^2): 15 power density (W/cm^2): 0.025 treatment time: 1 week (five sessions)	Co-intervention: soft cupping control group also received placebo (sham) laser
Vallone et al. (2014) Italy	Mean age of around 64 years	I:50 C:50	Laser diode: GaAlA Pulse mode: continuous wavelength (nm): 980 spot size (cm^2): 32 dose/point (J): 1,200 energy density (J/cm^2): 37.5 power density (W/cm^2): 0.625 treatment time: 3 weeks (nine sessions)	Co-intervention: exercise control group also received placebo laser
Shin et al. (2015) Republic of Korea	Mean age of around 46 years	I:28 C:26	Laser diode: Infrared 4-channel Pulse mode: pulsed wavelength (nm): 660 spot size (cm^2): 0.2 energy density (J/cm^2): 22.5 power density (W/cm^2): 0.0125 treatment time: 1 week (three sessions)	Co-intervention: cupping control group also received placebo laser
Gur et al. (2003) Turkey	Mean age of around 36 years; duration of back pain ~15 months and mean BMI of 25 kg/m^2	I:25 C:25	Laser diode: Gallium-arsenide Pulse mode: pulsed energy density (J/cm^2): 1 power density (W/cm^2): 0.0042 treatment time: 4 weeks (20 sessions)	Co-intervention: exercise
de Carvalho et al. (2016) Brazil	Mean age of around 40 years; duration of back pain ~14 months	I:18 C:13	Laser diode: AsGa laser Pulse mode: pulsed wavelength (nm): 904 energy density (J/cm^2): 0.23 power density (W/cm^2): 0.2375 treatment time: 15 days (15 sessions)	Co-intervention: exercise control group also received placebo laser
Tantawy et al. (2019) Saudi Arabia	Mean age of around 37 years; duration of back pain ~4 years and mean BMI of 27 kg/m^2	I:15 C:15	Laser diode: GaAlA Pulse mode: continuous wavelength (nm): 808 energy density (J/cm^2): 17 power density (W/cm^2): 0.1136 treatment time: 8 weeks (16 sessions)	Co-intervention: exercise
Hsieh and Lee (2014) Taiwan	Mean age of around 60 years; mean BMI of 24 kg/m^2	I:33 C:27	Laser diode: gallium-aluminum-arsenide diodes laser wavelength (nm): 890 energy density (J/cm^2): 83.2 power density (W/cm^2): 0.0347 treatment time: 2 weeks (six sessions)	Co-intervention: hot packs
Choi et al. (2017) Republic of Korea	Mean age of around 48 years; mean body weight of 63 kg	I:10 C:10	Energy density (J/cm^2): 13.8 treatment time: 4 weeks (12 sessions)	Co-intervention: physical therapy/exercise
Abdelbasset et al. (2020b) Saudi Arabia	Mean age of around 33 years; mean BMI of 26 kg/m^2 and mean duration of back pain of around 8 months	I:20 (each in high and low-dose group C:20	Laser diode: GaAlA (for low dose) and gallium-arsenide diode laser (for high dose) pulse mode: continuous wavelength (nm): 850 (low dose) and 1,064 (high dose) spot size (cm^2): 1.0 energy density (J/cm^2): 50 (in low dose) and 150 (in high dose) treatment time: 12 weeks (24 sessions)	Co-intervention: exercise
Abdelbasset et al. (2020a) Saudi Arabia	Mean age of around 39 years; majority males (>60%) and mean duration of back pain of around 7 months	I:18 C:17	Laser diode: Nd:YAG Pulse mode: pulsed wavelength (nm): 1,064 energy density (J/cm^2): 120-150 treatment time: 6 weeks (18 sessions)	Co-intervention: exercise control group also received placebo (sham) laser
Cheng et al. (2023) Taiwan	Mean age of around 34 years; mean BMI of 25 kg/m^2	I:53 C:52	Laser diode: gallium-aluminum-arsenide diodes laser wavelength (nm): >600 energy density (J/cm^2): 4.5 treatment time: 2 weeks (10 sessions)	Control group received standard postpartum care

(*continued on next page*)

Table 1 (*continued*)

Author (Publication Year) and Place of Study	Subject Characteristics	Sample Size	Details of Laser Acupuncture	Co-Intervention/ Control Group
Yang et al. (2023) Taiwan	Mean age of around 37 years; mean BMI of 24 kg/m^2; low back pain for more than 12 weeks (in 60%)	I:38 C:38	Laser diode: gallium-aluminum-arsenide diodes laser wavelength (nm): 810 power density (W/cm^2): 1.12 treatment time: 4 weeks (eight sessions)	Control group received placebo (sham) laser
Lin et al. (2017) Taiwan	Mean age of around 63 years and mean BMI of around 23 kg/m^2	I:20 C:20	Laser diode: LA-400 Pulse mode: pulsed wavelength (nm): 808 energy density (J/cm^2): 15 treatment time: 5 days (five sessions)	Co-intervention: soft cupping control group also received placebo (sham) laser
Kim et al. (2022) Republic of Korea	Mean age of around 55 years; majority females (>60%) and mean weight of around 65 kg	I:15 C:15	Laser diode: GaAlA wavelength (nm): 830 dose/point (J): 12 energy density (J/cm^2): 38,216.5 power density (W/cm^2): 63.69 treatment time: 4 weeks (eight sessions)	Control group received placebo (sham) laser
Ammar (2015) Egypt	Mean age of around 40 years; mean duration of pain around 5 years and mean BMI of 26 kg/m^2	I: 35 C: 35	Wavelength (nm): 850 dose/point (J): 5 treatment time: 6 weeks (12 sessions)	Co-intervention: exercise control group also received hot packs

Chen et al. (2022) found that in terms of pain relief, laser therapy had a short-term effect in patients with nonspecific chronic LBP (Chen et al., 2022). The review also highlighted that there is not enough data to determine whether there is a long-term benefit of laser therapy, and what is the ideal dose or type of laser (Chen et al., 2022).

One possible explanation for the observed short-term effect is that the stimulation of specific acupoints by LA could activate endorphin release and modulate pain perception (Chon et al., 2019; Zeng et al., 2018). Additionally, the increase in blood flow and circulation caused by the laser may contribute to temporary pain reduction (Lim et al., 2018). Nonetheless, the limited duration of the

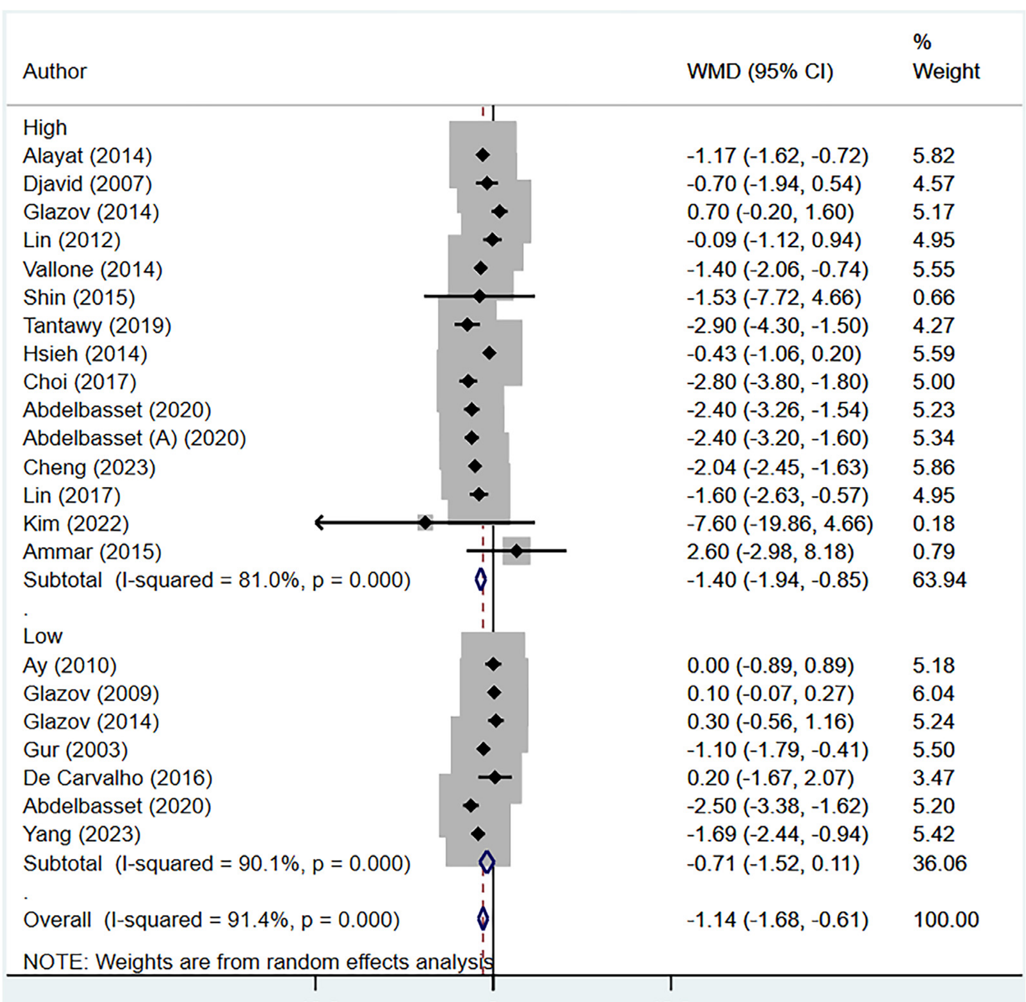

Figure 3. Pain scores assessed immediately after completion of laser acupuncture, compared to control group.

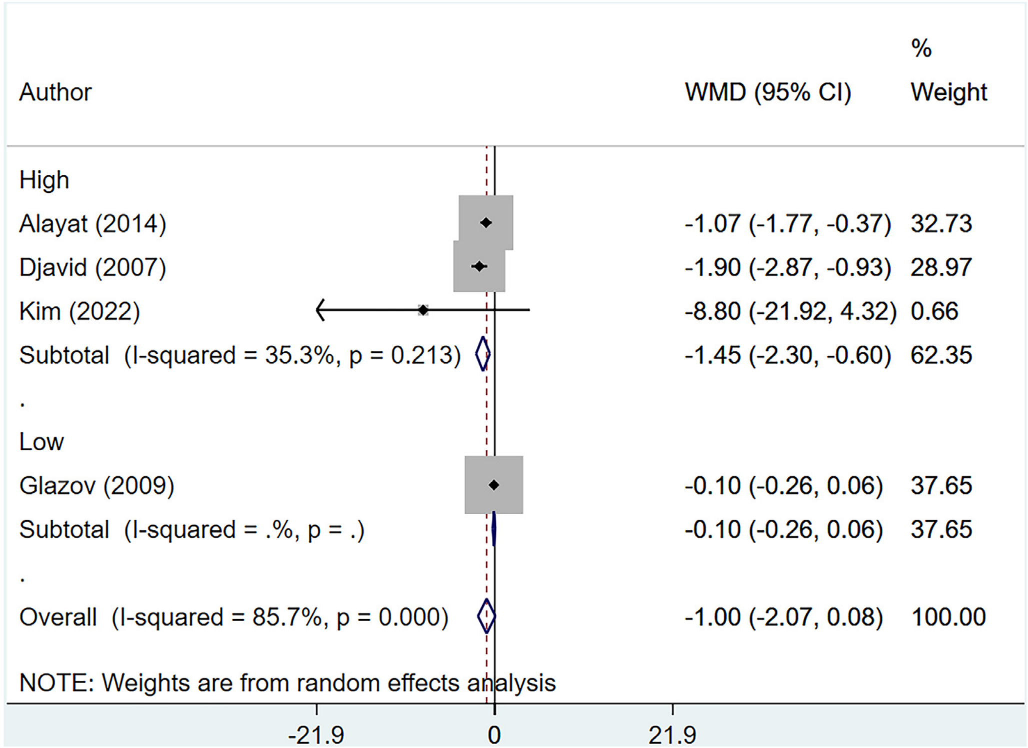

Figure 4. Pain scores assessed at 4-8 weeks (short-term follow-up) after completion of laser acupuncture, compared to control group.

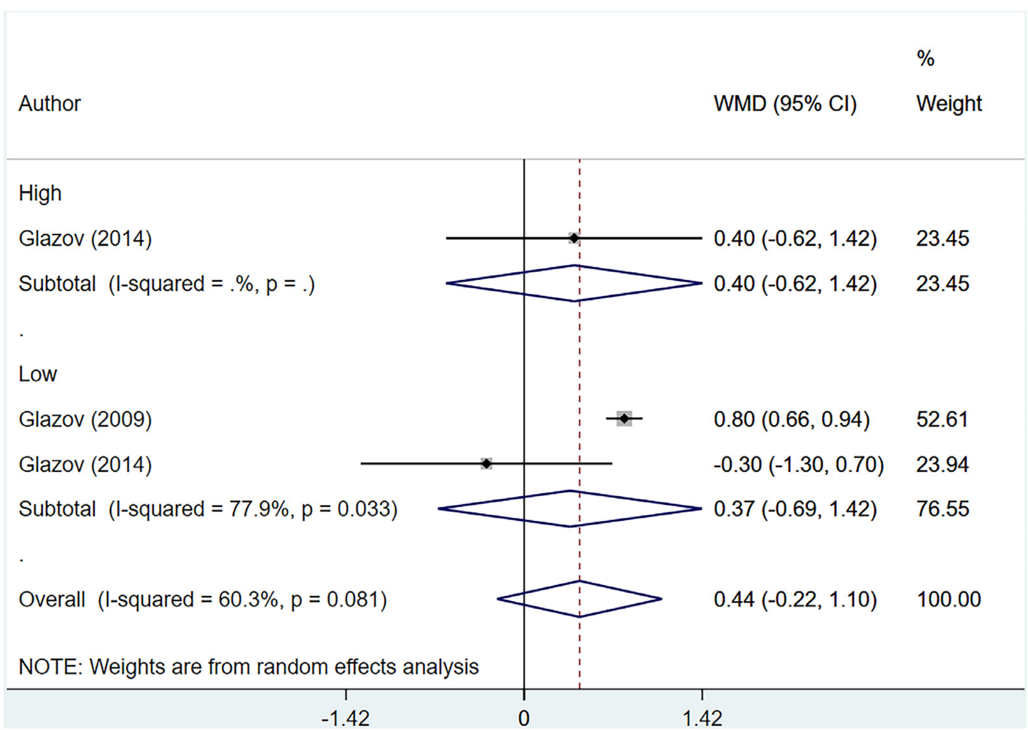

Figure 5. Pain scores assessed at 12 months (long-term follow-up) after completion of laser acupuncture, compared to control group.

effect highlights the need for further investigation into the underlying mechanisms of the pain-relieving properties of LA. The results of our meta-analysis revealed that high-dose LA was associated with greater pain reduction compared to low-dose LA. One plausible explanation for this observation is the deeper location of anatomical structures in the low back area, which may necessitate a higher dose of laser irradiation for adequate penetration and therapeutic effect.

Our meta-analysis presents several strengths when compared to previous reviews on the efficacy of LA for LBP. We had a clear inclusion criterion which allowed only RCTs focusing on chronic LBP in adults to be included in the review. This ensured methodological rigor and minimized bias in study selection. Moreover, our study concentrated on contemporary research conducted within the last two decades, offering a more up-to-date and relevant assessment of LA effectiveness. The lack of sustained effectiveness during later follow-up assessments is an important finding that suggests that the benefits of LA in reducing LBP may not be long-lasting. Several factors could contribute to this phenomenon, including the absence of maintenance treatments, the presence of underlying chronic conditions, and individual variability in response to LA.

While this meta-analysis provides valuable insights into the effectiveness of LA for LBP, some limitations need to be acknowledged. First, all included studies had relatively small sample sizes, which may have affected the statistical power of the analysis. Additionally, the quality of the included studies varied, and some studies lacked sufficient methodological rigor, potentially introducing bias into the meta-analysis. We found a high degree of heterogeneity for the outcomes in our study. This could be due to the possible differences in the treatment protocols, laser parameters, and patient characteristics. Additionally, we encountered challenges in identifying a clear dosage effect or determining whether a high dose is more advantageous than a low-dose laser therapy. The therapeutic sessions reported in the included studies were also inconsistent, adding to the complexity of the analysis.

Relevance of the Findings to Nursing Practice

The findings of the review on the efficacy of LA for the treatment of chronic LBP hold particular relevance to nursing practice. Nurses often play a crucial role in pain management, and understanding the temporal dynamics of LA effects is essential for informed patient care. The observed immediate pain relief posttreatment with LA highlights a potential adjunctive intervention that nurses can incorporate into pain management strategies for patients with chronic LBP. However, the crucial insight that the effect does not appear to be sustained on later follow-up assessments is equally important for nursing practice. Nurses can use this information to guide patient education, managing expectations, and developing comprehensive, individualized pain management plans. Incorporating LA as part of a broader, multi-modal approach to pain management aligns with nursing's holistic care perspective, emphasizing not only immediate relief but also addressing the long-term needs and well-being of patients experiencing chronic LBP.

Conclusion

This meta-analysis shows that in patients with LBP, LA may provide a short-term transient decrease in pain scores immediately after the treatment. Our findings underscore the need for additional high-quality RCTs with larger sample sizes and standardized treatment protocols to gain a full understanding of the efficacy and long-term effects of LA for LBP. Healthcare practitioners, including nursing personnel, should consider these findings when recommending LA as a treatment option, and patients should be informed about its potential limitations in providing lasting pain relief. Nursing personnel, in particular, should be trained to manage patient expectations, emphasizing its role within a broader, long-term approach to chronic pain care.

Declaration of competing interest

The authors declare that they have no conflict of interest.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.pmn.2024.05.001.

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